

Targeting the top 20% could materially improve telephone-campaign productivity

Recommendation

Replace broad customer selection with a **pre-campaign logistic ranking**, then pilot the policy before full rollout.

49.7%

top 20% conversion

30.8%

test-period random baseline

+311

incremental subscriptions

- ▶ In the ordered test period, the top 20% of scored customers converted at **49.7%** versus **30.8%** under random selection.
- ▶ This is a **1.61×** lift across **1,648** calls.
- ▶ Use the ranking as a controlled pilot, not as immediate full-scale automation.

The decision is whom to call first – not who appears likely after the call

- ▶ **Objective:** maximize subscriptions within fixed telephone capacity.
- ▶ **Scoring moment:** before current-campaign contact.
- ▶ **Valid inputs:** customer profile, financial relationship, prior-campaign history, and known macro context.
- ▶ **Primary metrics:** conversion rate and lift at capacity cuts.

softblue **Pre-campaign**
customer + prior history
USED

Current campaign
campaign/contact activity
EXCLUDED

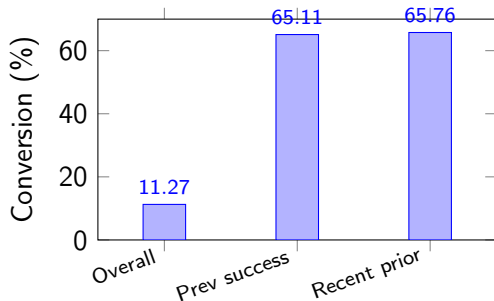
Post-call
duration + outcome
EXCLUDED

Excluded from strict model

- ▶ duration: known after the call.
- ▶ campaign: generated during current campaign.
- ▶ contact, month, day_of_week: current scheduling/channel fields.

Prior response history separates high- and low-conversion groups, but conditions change over time

- ▶ Overall dataset conversion: **11.27%**.
- ▶ Previous campaign success: **65.11%** conversion.
- ▶ Previous contact within 0–7 days: **65.76%** conversion.
- ▶ Ordered-data quintiles ranged from **3.14%** to **30.84%** conversion.
- ▶ duration bins ranged from near-zero to **48.6%**, demonstrating severe leakage.

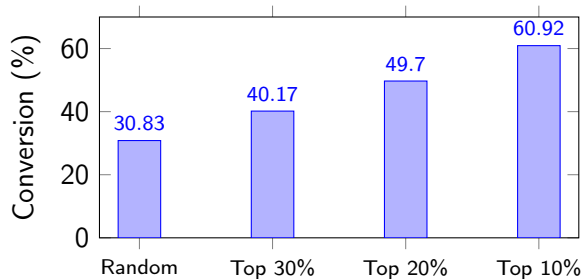


Interpretation

The data contains exploitable heterogeneity, but substantial temporal drift requires forward-style validation.

The simpler model produces the strongest operational ranking

Model	ROC-AUC	PR-AUC	Top-10% lift	Top-20% lift	Selected
Logistic	0.559	0.418	1.98×	1.61×	Yes
Gradient boosting	0.588	0.394	1.45×	1.57×	No



Main takeaway

Gradient boosting modestly improved ROC-AUC, but underperformed logistic regression on the most relevant ranking metrics.

At the top 20% cutoff, logistic ranking yielded **819 conversions** across **1,648 calls**, or about **311 more** than random selection in the same test period.

Deploy through a controlled pilot, not immediate full-scale automation

Pilot design

- ➊ Score eligible customers before campaign launch.
- ➋ Randomize customers near or across the cutoff:
 - scored targeting group;
 - business-as-usual or random control.
- ➌ Measure conversions per call, deposits and margin, agent time, and complaints or opt-outs.
- ➍ Retrain and recalibrate as campaign conditions change.

Risks and limitations

- ▶ Prediction is not causal uplift.
- ▶ No stable customer identifier.
- ▶ Source order is only approximate chronology.
- ▶ Train conversion was 6.37%; test conversion was 30.83%.
- ▶ Historical campaign selection may bias labels.
- ▶ No deposit amount, margin, or call-cost fields.
- ▶ Probabilities are not calibrated enough for direct economic optimization.